Unggi Lee

Real-time systems · DNN inference offloading · Image-adaptive compression

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EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST) — Feb. 2026 - Feb. 2028 (Expected)

M.S. in Computer Science — *Daegu, S.Korea*

- Real-Time Computing Lab (RTCL)

Daegu Gyeongbuk Institute of Science and Technology (DGIST) — Feb. 2020 - Feb. 2026

B.S. in Electrical Engineering and Computer Science — *Daegu, S.Korea*

- Total GPA of 3.82 / 4.30

RESEARCH EXPERIENCE

Real-Time Computing Lab (RTCL), DGIST — Feb. 2026 - Present

Graduate Research Assistant — *Daegu, S.Korea*

- Real-time systems, with a focus on DNN inference offloading between edge devices and servers under latency constraints.
- Image-adaptive JPEG compression: per-image quality factor and quantization table optimization, so that offloaded inference holds up under variable network bandwidth.

PROJECT EXPERIENCE

WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation — 2025

Reinforcement Learning course project, published at ICTC 2025 — *Daegu, S.Korea*

- A model-free PPO framework that assembles top-bottom outfit pairs from a natural-language style prompt, rewarded by CLIP similarity fused with a learned proxy for human preference.

Noise Map-Driven Dynamic Frame Processing for Improved Object Detection in Low-Light Conditions — 2024

Undergraduate Group Research Program (UGRP), Team of Four — *Daegu, S.Korea*

- Placed an enhancement front-end before YOLO — Zero-DCE for low-light enhancement, CBDNet for noise-map estimation, and MFDNet for denoising — integrated directly into the detector's data loader.
- Held the pipeline to real time by exploiting temporal redundancy between frames, processing one frame in five at 20 FPS.
- Improved mAP by roughly 1% over the unenhanced baseline on BDD100k.

PUBLICATIONS

WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation — Oct. 2025

Minseok Oh, **Unggi Lee***, Hyunmin Kim, Dooseok Lee* — *ICTC 2025*

- 2025 16th International Conference on Information and Communication Technology Convergence (ICTC) — PDF / IEEE Xplore

INTERNATIONAL EXPERIENCE

University of California, Los Angeles (UCLA) — Jun. - Aug. 2023

Summer Session — *Los Angeles, CA, USA*

SKILLS

Programming	C, C++, Python
Deep Learning	PyTorch
Computer Vision	OpenCV
Tools	Docker, Linux, Git